M. Markham, D.J. Twitchen

C. Abellán, W. Amaya,
V. Pruneri, M.W. Mitchell

B. Hensen, H. Bernien, A.E. Dréau,
A. Reiserer, N. Kalb, M.S. Blok, J. Ruitenberg,
R.F.L. Vermeulen, R.N. Schouten, D. Elkouss,
S. Wehner, T.H. Taminiau, R. Hanson

Experimental loophole-free violation of a Bell inequality using entangled electron spins separated by 1.3 km

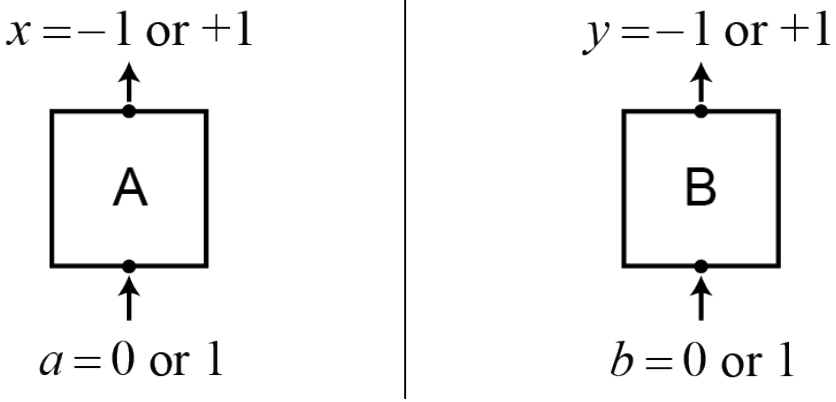
arXiv:1508.05949

- Introduction
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Bell's theorem





$$S = \left|\langle x \cdot y\rangle_{(0,0)} + \langle x \cdot y\rangle_{(0,1)} + \langle x \cdot y\rangle_{(1,0)} - \langle x \cdot y\rangle_{(1,1)}\right|$$

1. Communication is limited to speed of light
2. Settings *a,b* can be freely chosen
3. **Local hidden variable model (LHVM)** governs the experiment

$$\Rightarrow \quad S \leq 2$$

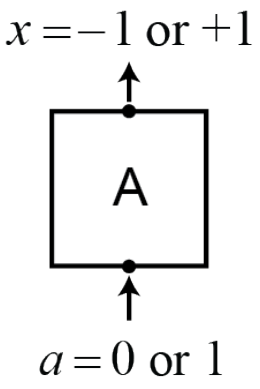
3'. **Quantum theory** predicts

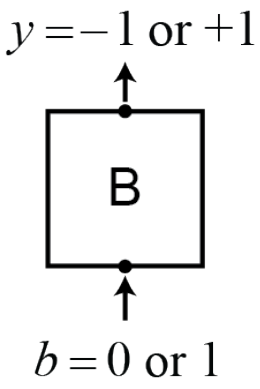
$$\Rightarrow \quad S \leq 2\sqrt{2}$$

Bell's game

Game score

$$C=\begin{cases} 1 & \text{if } (-1)^{ab}xy = 1 \\ 0 & \text{else} \end{cases}$$





1. Communication is limited to speed of light
2. Settings a,b can be freely chosen uniformly at random
3. **Local hidden variable model (LHVM)** governs the experiment

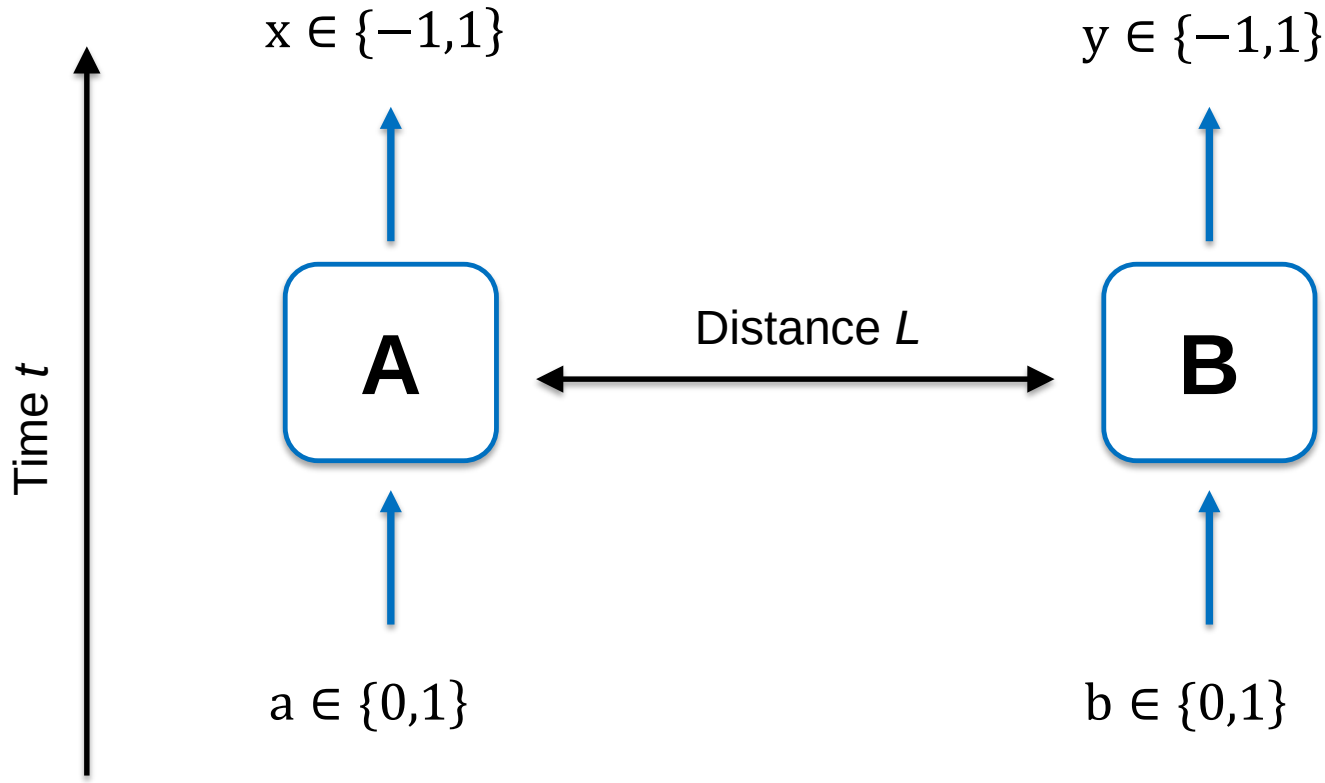
$\Rightarrow$ $P(C = 1) \leq 3/4$

3'. **Quantum theory** predicts

$\Rightarrow$ $P(C = 1) \leq (\sqrt{2}+2)/4 \approx 0.85$

Unless…

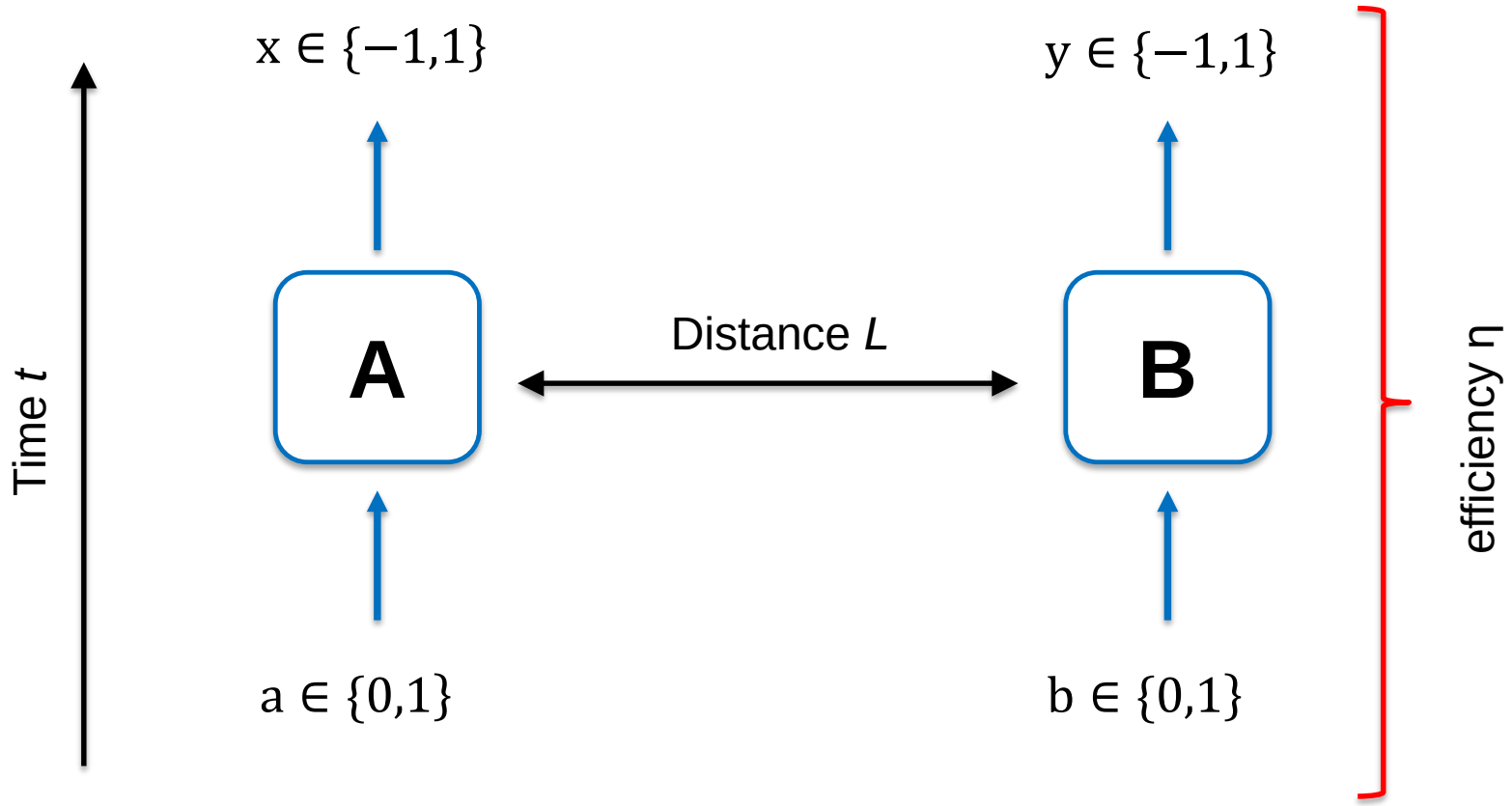
Locality loophole



- If L is so short that Alice can signal to Bob before y is produced then y=f(a,b)

- Strategy with winning probability one: x=1, $y = (-1)^{ab}$

Detection loophole



- Outcomes are not always conclusive → data discarded

- Strategy with winning probability one: if a=0 output x=1, if b=0 output y=1 else no output

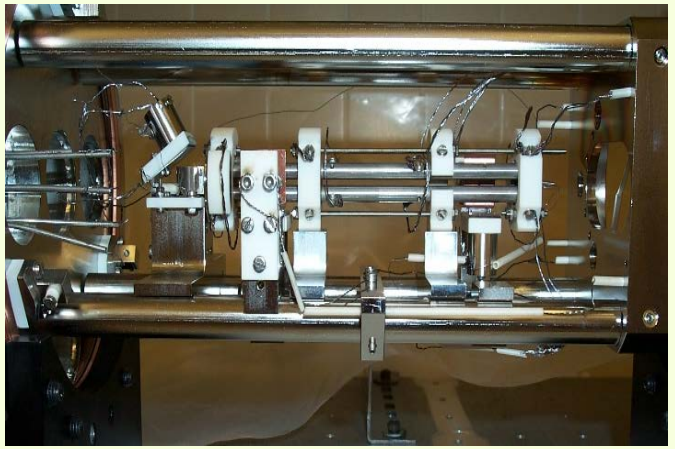
Bell experiments

Locality loophole addressed (photons)



Aspect (1982), Weihs (1998), Tittel (1999)

Detection loophole closed (ions, atoms, superconducting qubits, photons)



Rowe (2001), Matsukevich (2008), Ansmann (2009), Giustina (2013), Christensen (2013)

Detection loophole closed AND locality loophole addressed = **“loophole-free” Bell test**

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Setup for closing the loopholes

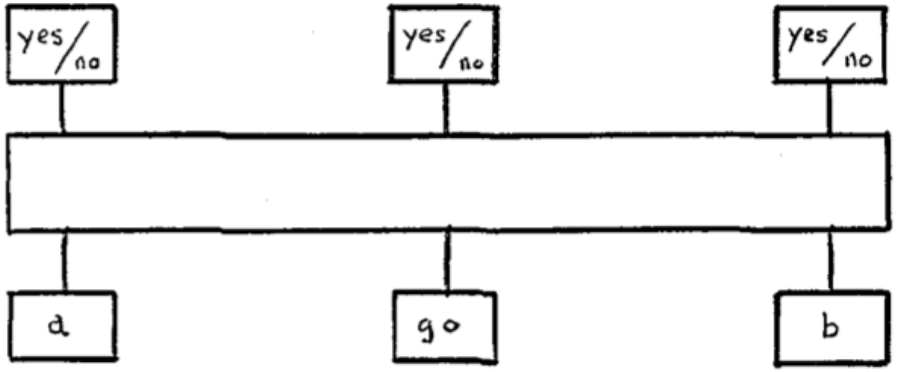
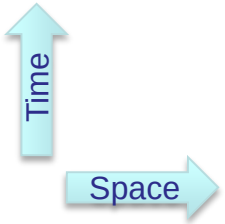
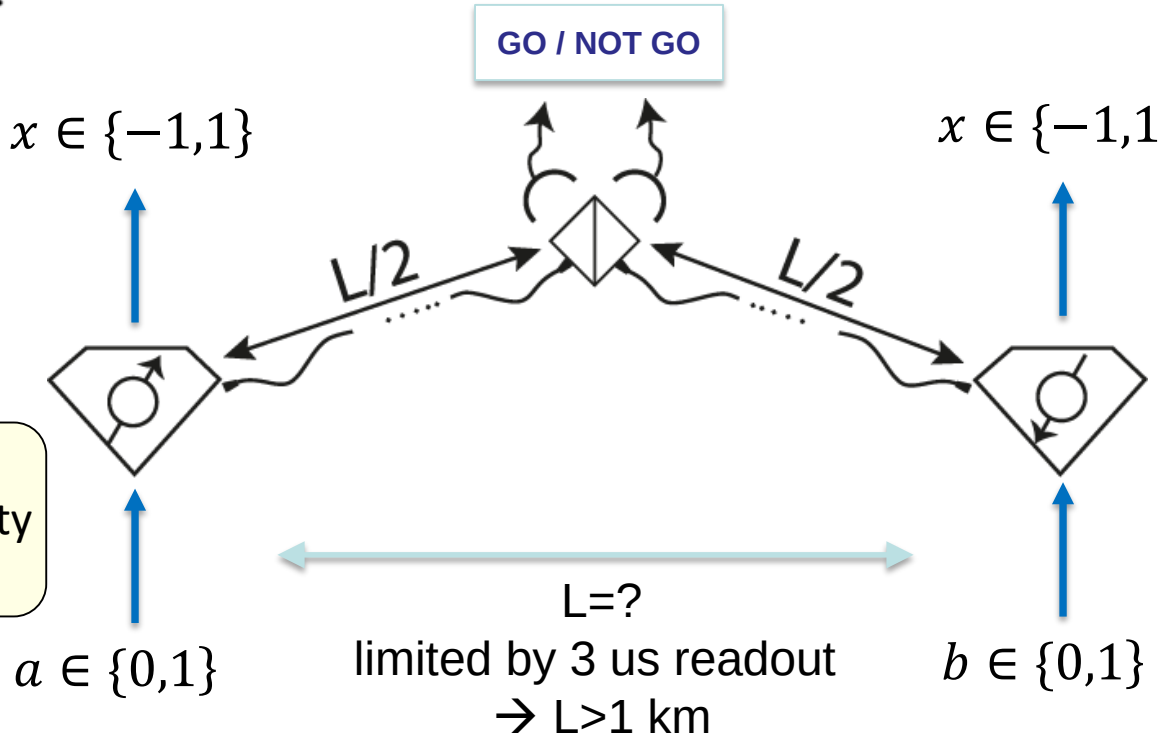


Fig. 7.- General EPR set-up, with three inputs below and three outputs above.

Bell, J. Phys. (Paris) Colloq. C2, 41 (1981).

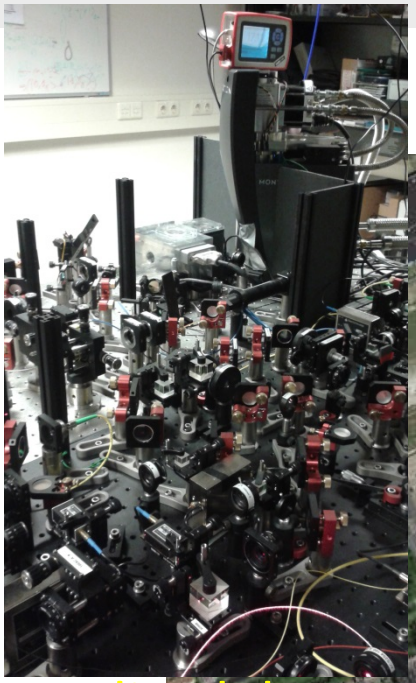




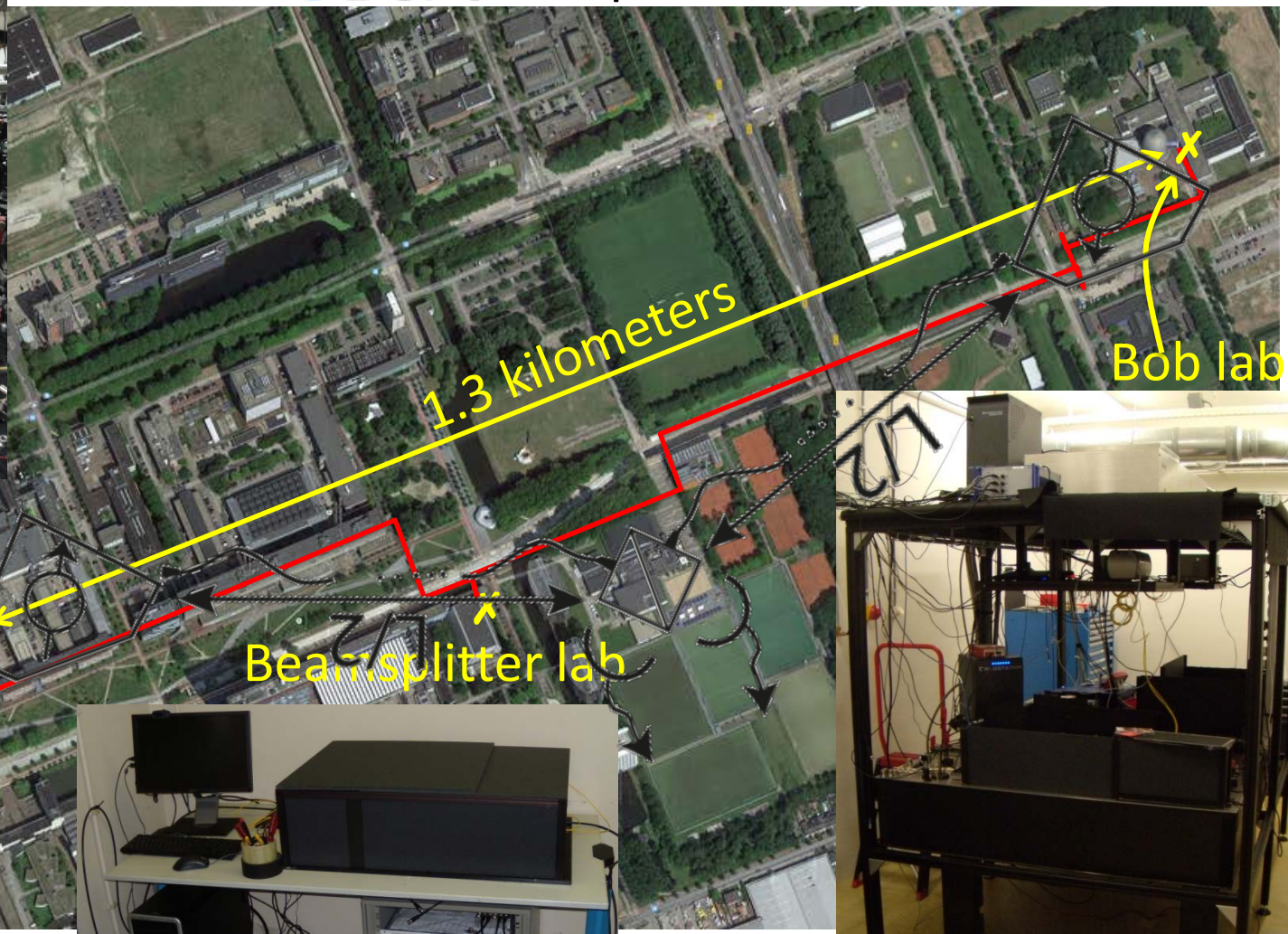
Matter qubits: "easy" high fidelity readout

Phys. Rev. Lett. 91, 110405 (2003).

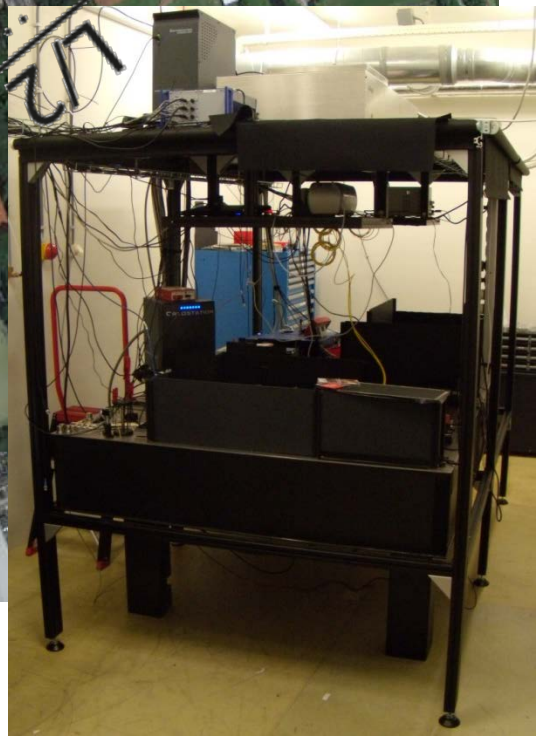
Loophole-free Bell test



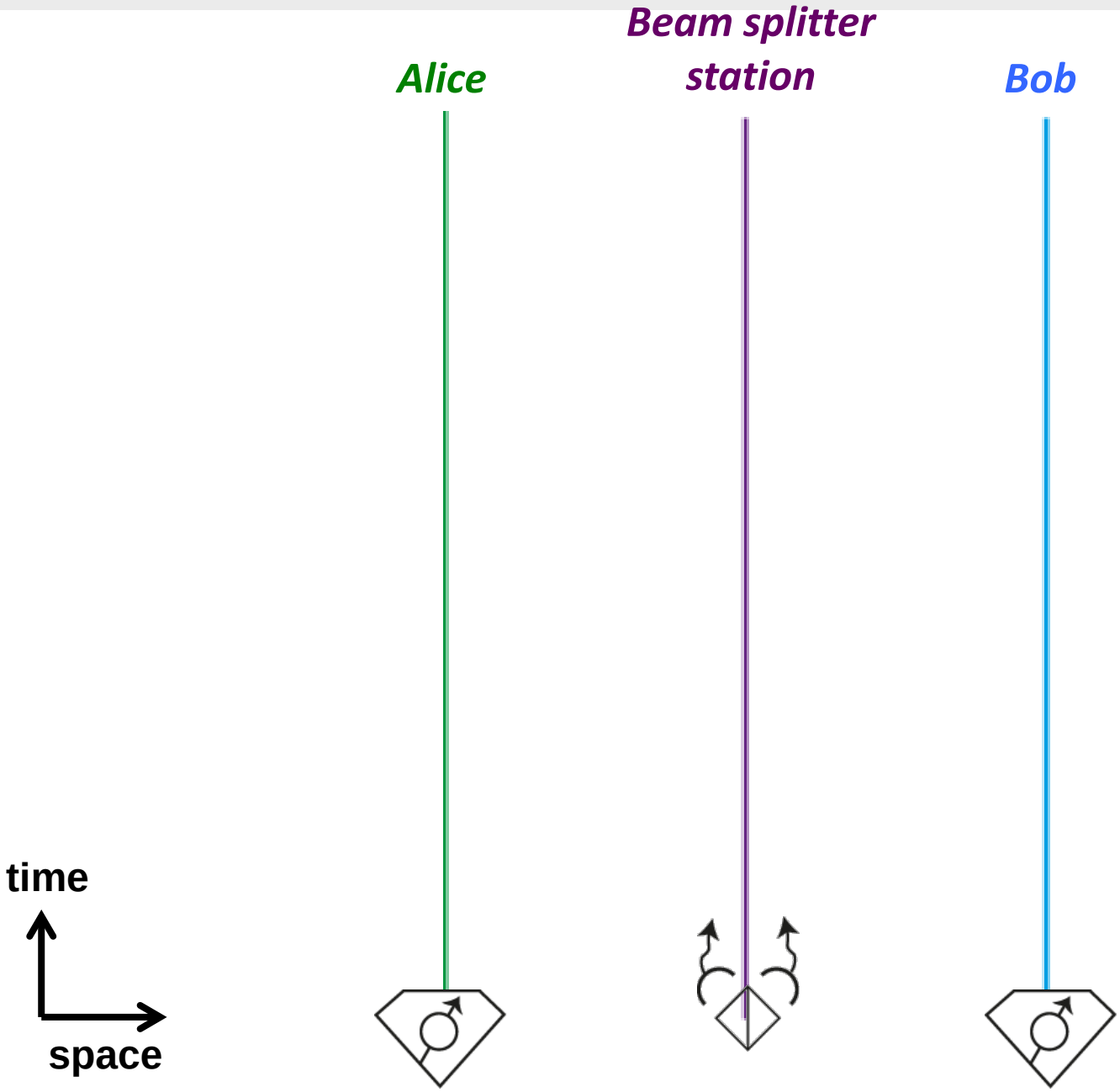
TU Delft campus







Experimental scheme

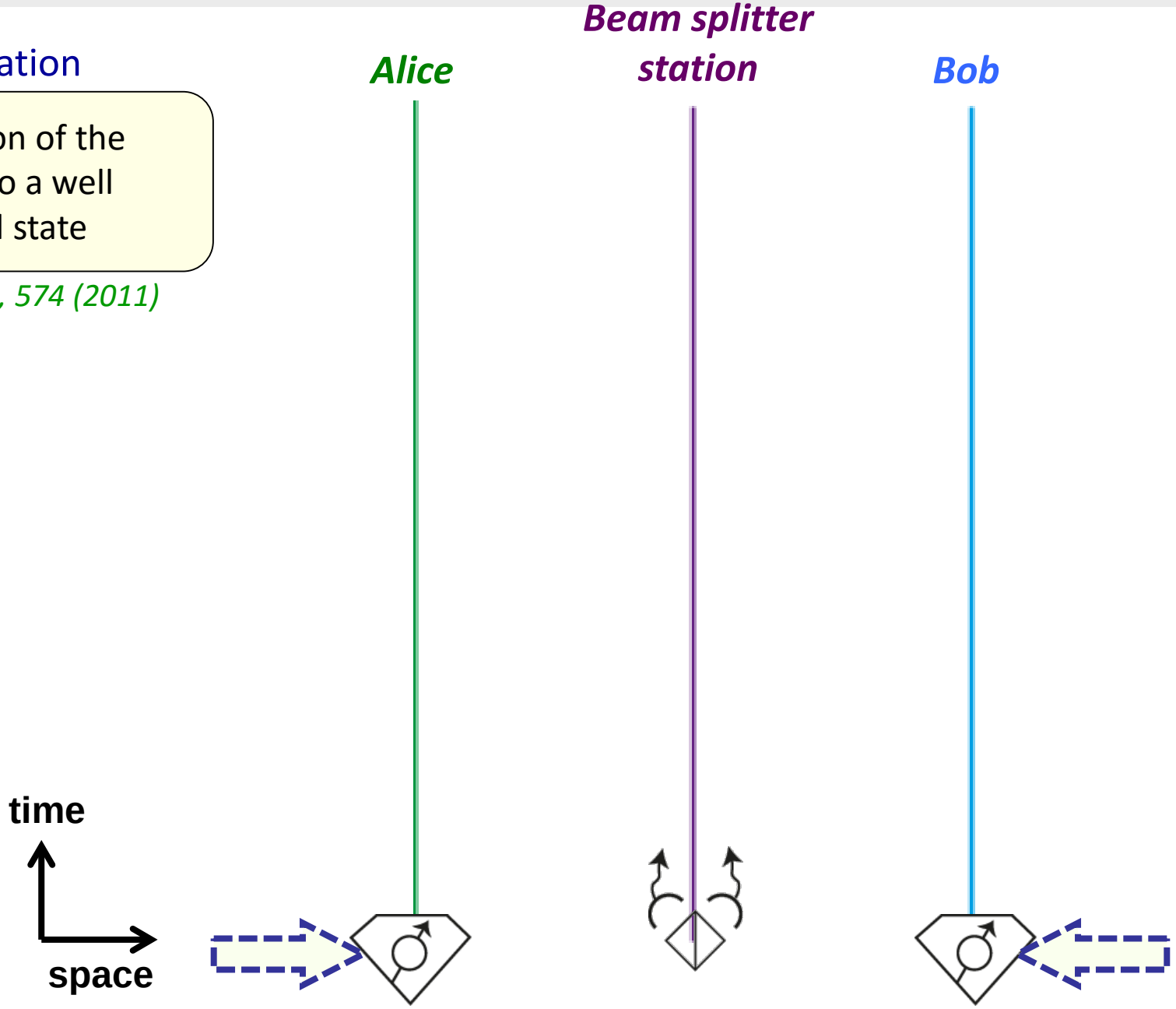


Experimental scheme

Initialization

Initialization of the qubits into a well defined state

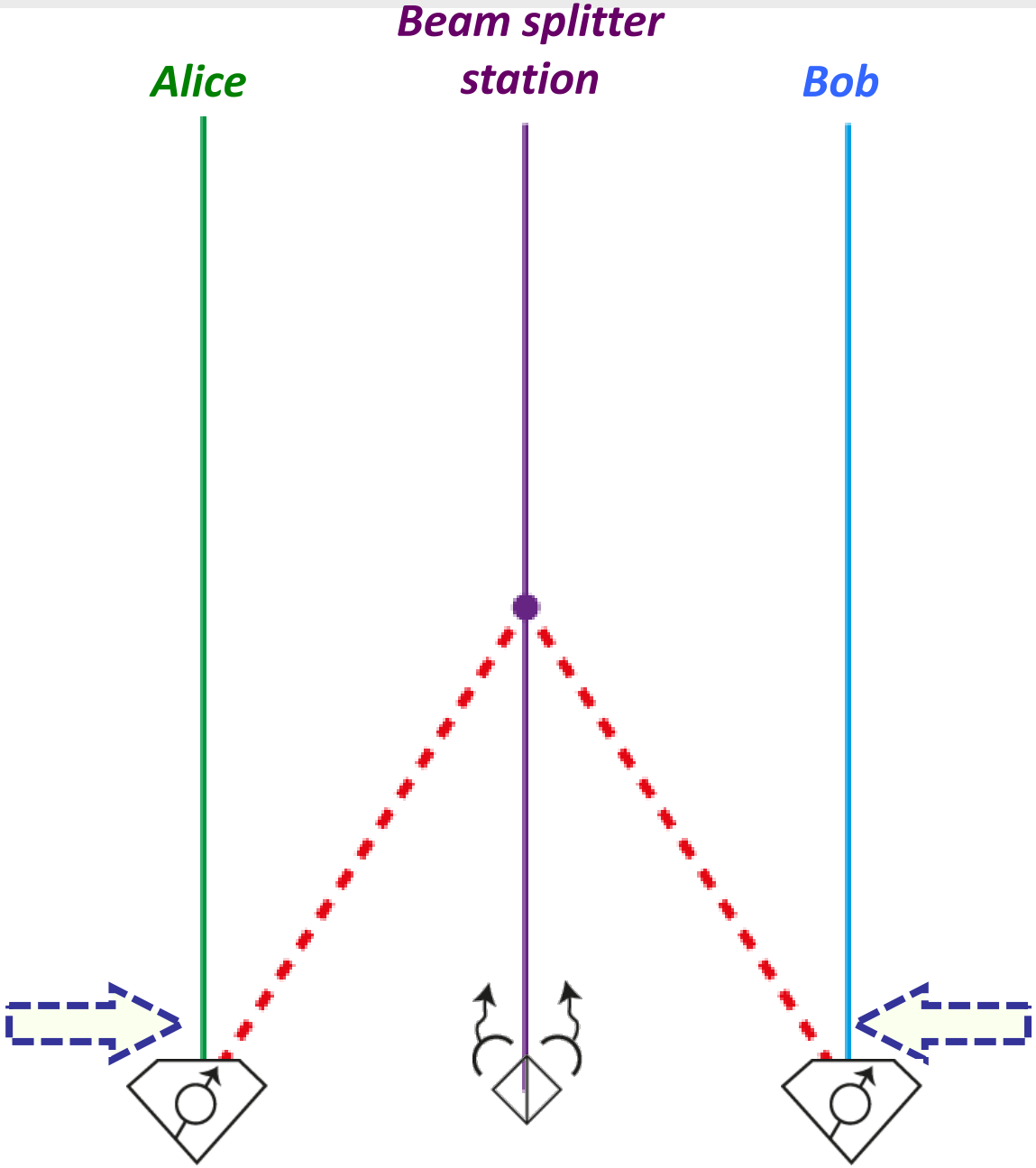
Nature 477, 574 (2011)



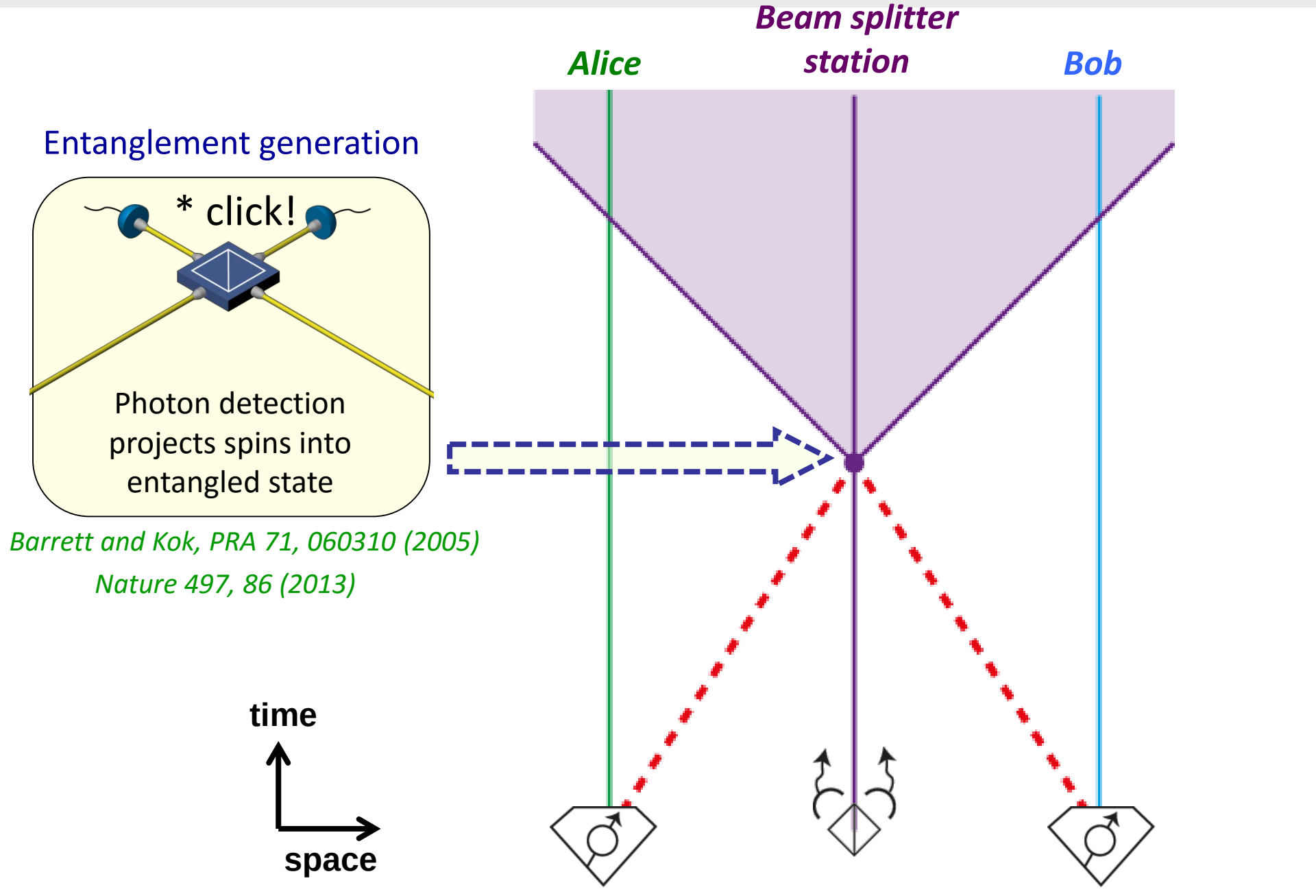
Experimental scheme

Spin-photon entanglement

Create spin-photon entanglement on both sides.
Send photons to beam splitter.

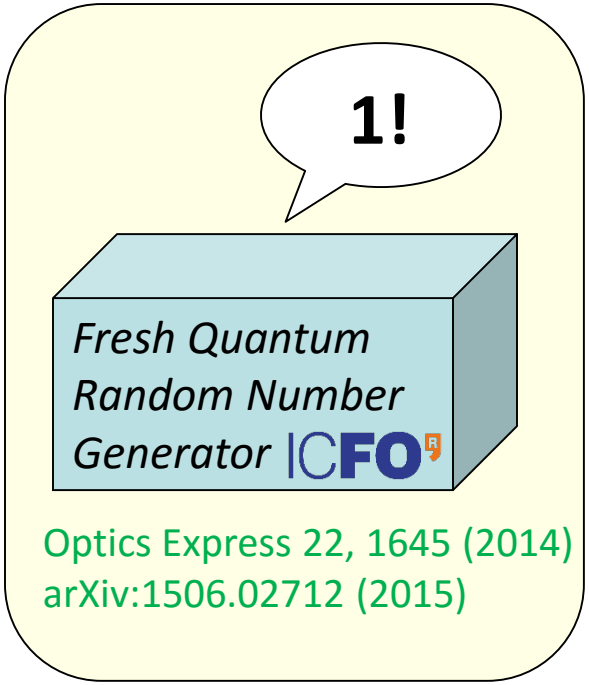


Experimental scheme

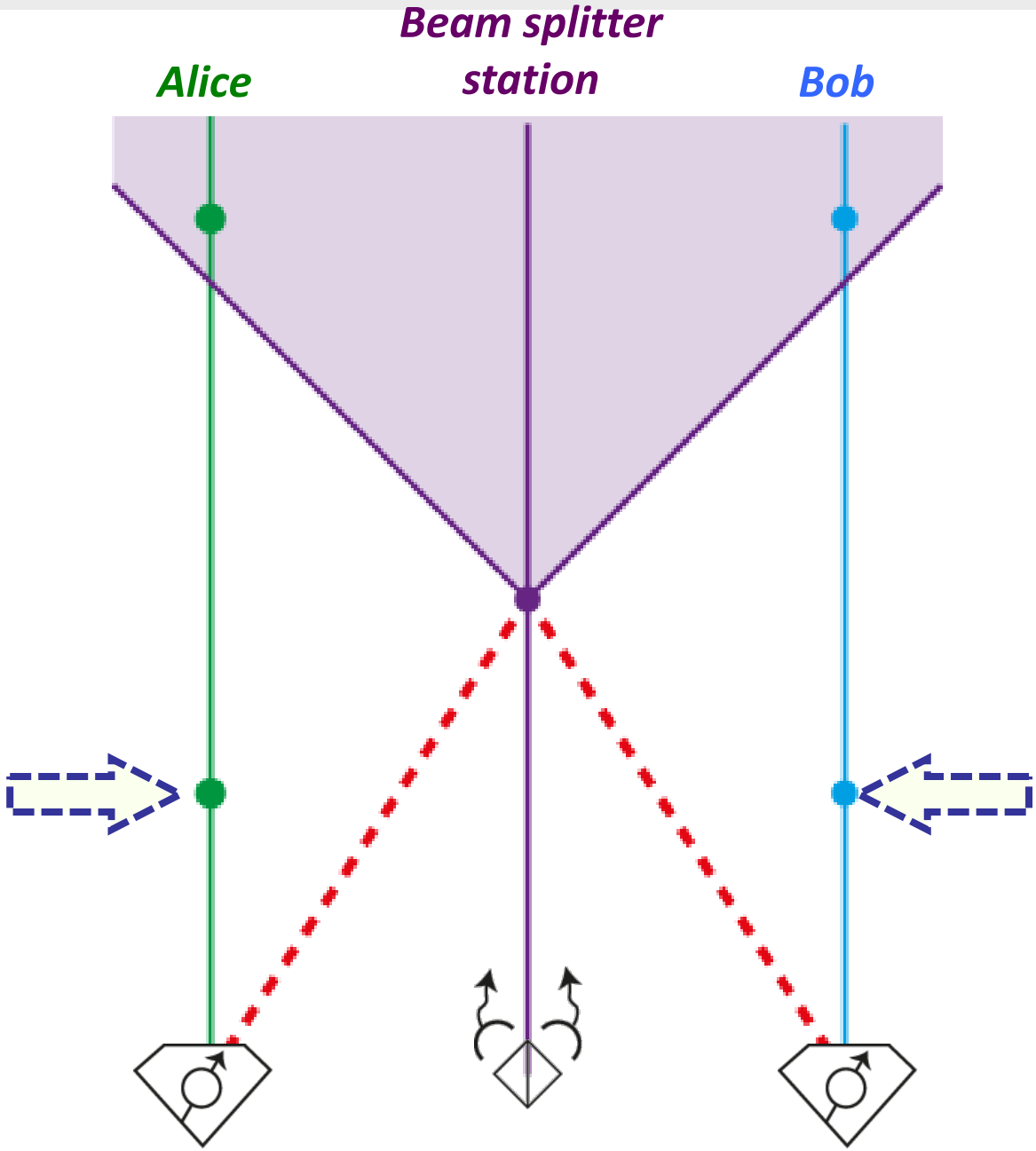


Experimental scheme

Random basis choice



Optics Express 22, 1645 (2014)
arXiv:1506.02712 (2015)



Experimental scheme

Readout

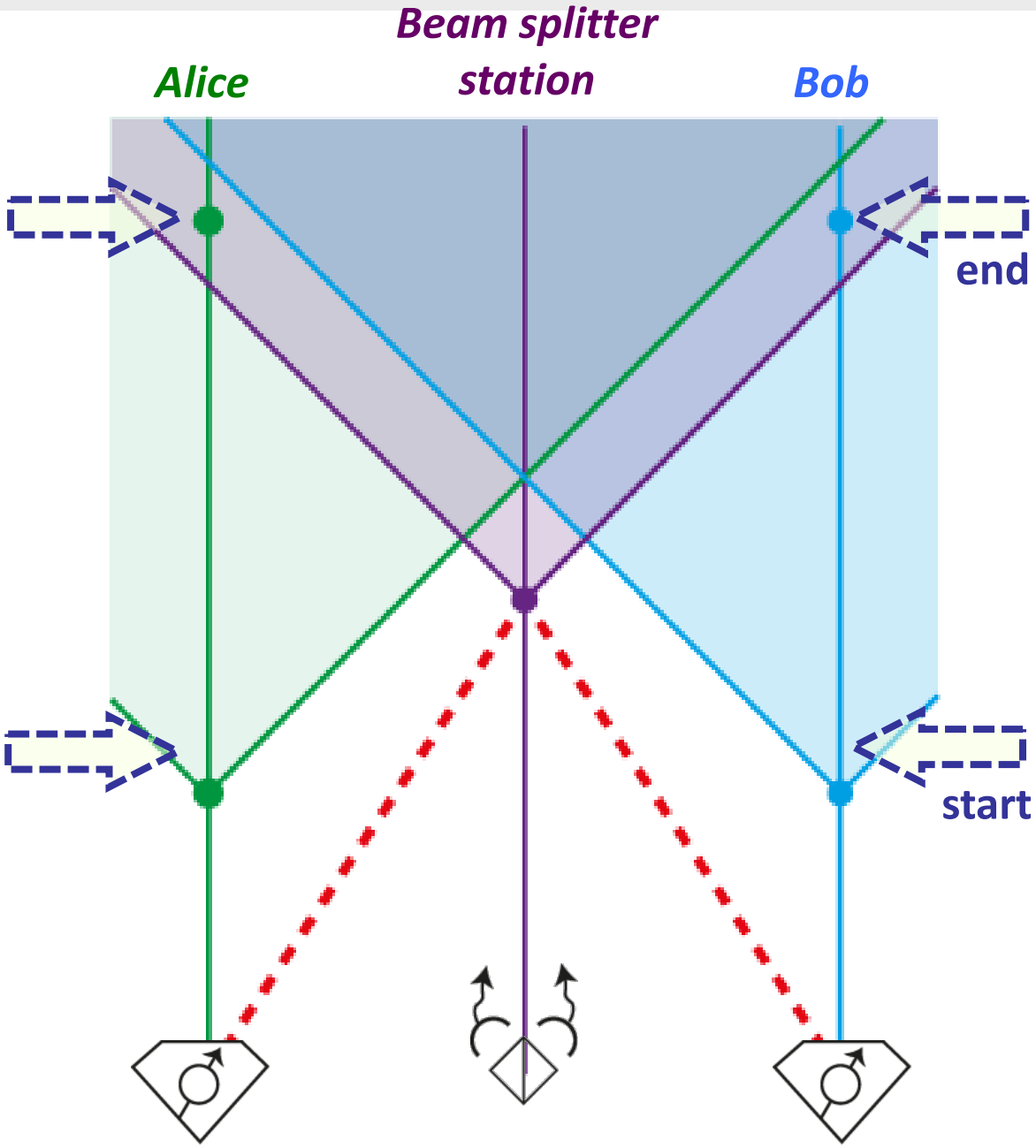
Distance such that signaling is not possible:

Locality loophole addressed

Efficient Readout:

Detection loophole closed

Nature 477, 574 (2011)



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Statistical analysis

LHVM imply limitations in probability or expectation

Finite number of events

Hypothesis test

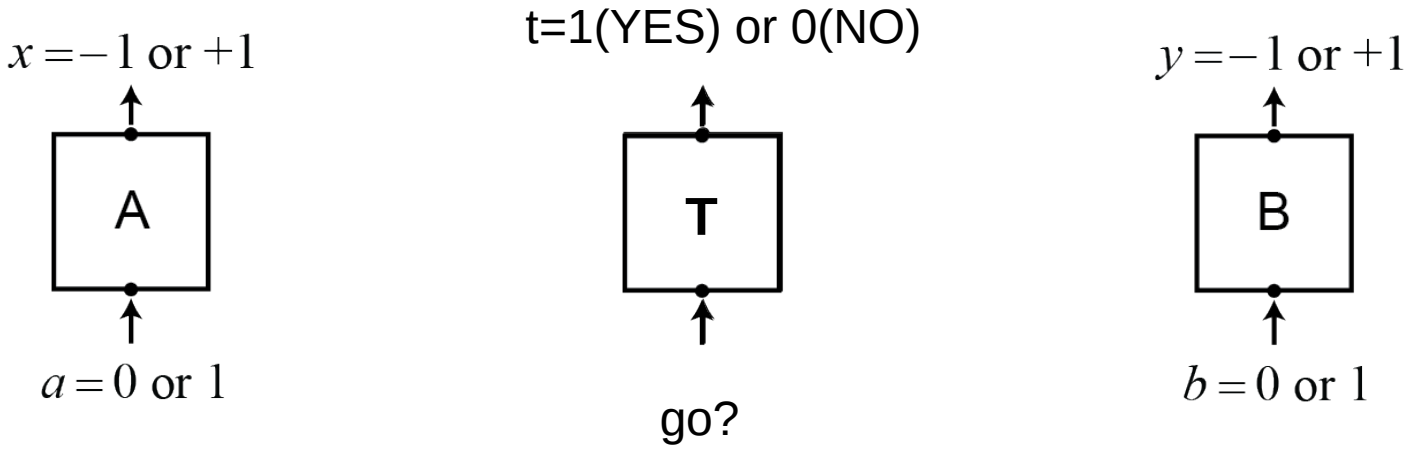
- **Null hypothesis**: the experiment was governed by a LHVM
- *p-value: Probability of data at least as extreme as that observed given a LHVM maximized over all LHVMs.*
- p-value is **not** the probability that our experiment is ruled by a LHVM

Rigorous bound

- No unfounded assumptions: i.i.d.
- No approximations: Gaussian

Gill, Found. of Prob. and Phy., 179-206 (2003)
Zhang, Glancy, Knill, PRA 84 062118 (2011)
Bierhorst, Found. Phys. 44, 736-761 (2014)

Challenges for modelling the experiment



$$C=\begin{cases} 1 & \text{if } t(-1)^{ab}xy = 1 \\ 0 & \text{else} \end{cases}$$

Local hidden variable model (LHVM) governs the experiment:

1. Setting choices are independent conditioned on the history
2. **Inputs have a bias from uniform τ (10^{-5})**
3. Settings and outcomes at both sites are conditionally independent on the history
4. **Arbitrary statistics at heralding station**

$$\Rightarrow \quad P(C = 1) \leq 3/4+3(\tau + \tau^2)$$

Upper bound on the p-value

Experimental Data

- n = number of GO rounds
- k = number of wins
- τ = bias of the RNGs

$$\textbf{p-value} := \max_{LHVM} P(at\ least\ k\ wins\ in\ n\ events \mid LHVM)$$

$$\leq \sum_{i=k}^{n} \binom{n}{k} \xi^k (1-\xi)^{n-k}$$

Remarks

- Upper bound is the tail of an i.i.d. distribution
- $\xi = 3/4 + 3(\tau + \tau^2)$

Elkouss, Wehner in preparation (2015)

Evaluation of the experiment

Gathered data

- 1 event per hour
- 245 events over more than 220 hours
- 196 “wins” over the 245 events

How do we interpret it?

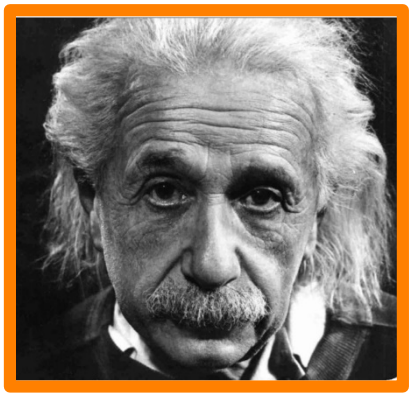
Hypothesis test

- **Null hypothesis:** the experiment was governed by a LHVM
- **p-value**: probability of data at least as extreme as that observed given a LHVM maximized over all LHVMs.

$$\text{p-value} := \max_{LHVM} P(at\ least\ 196\ wins\ in\ 245\ events \mid LHVM) \leq 0.039$$

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Conclusions



First loophole-free Bell inequality violation

arXiv:1508.05949 (2015)

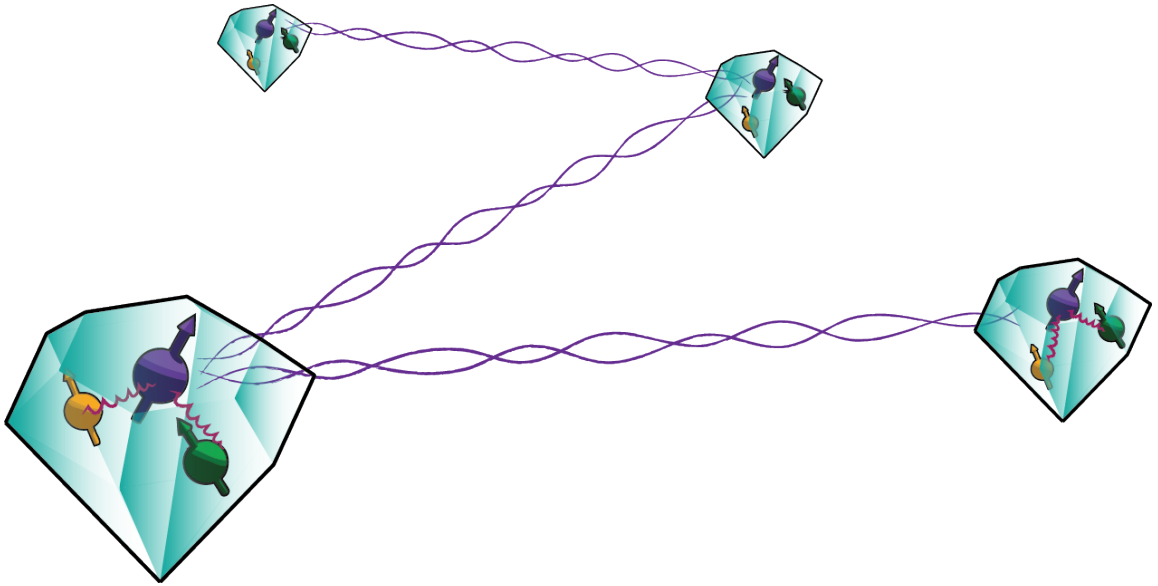


Next:

- Independent confirmation
- Proof-of-principle of device-independent security

…

- Quantum networks



Thank you!



FOM NWO erc S³NANO

どうもありがとうございます！



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